

DBMS – SELLER AND INVENTORY MANAGEMENT

Project Title: Gaming-Accessories E-Commerce Order Management System

1. Database Design

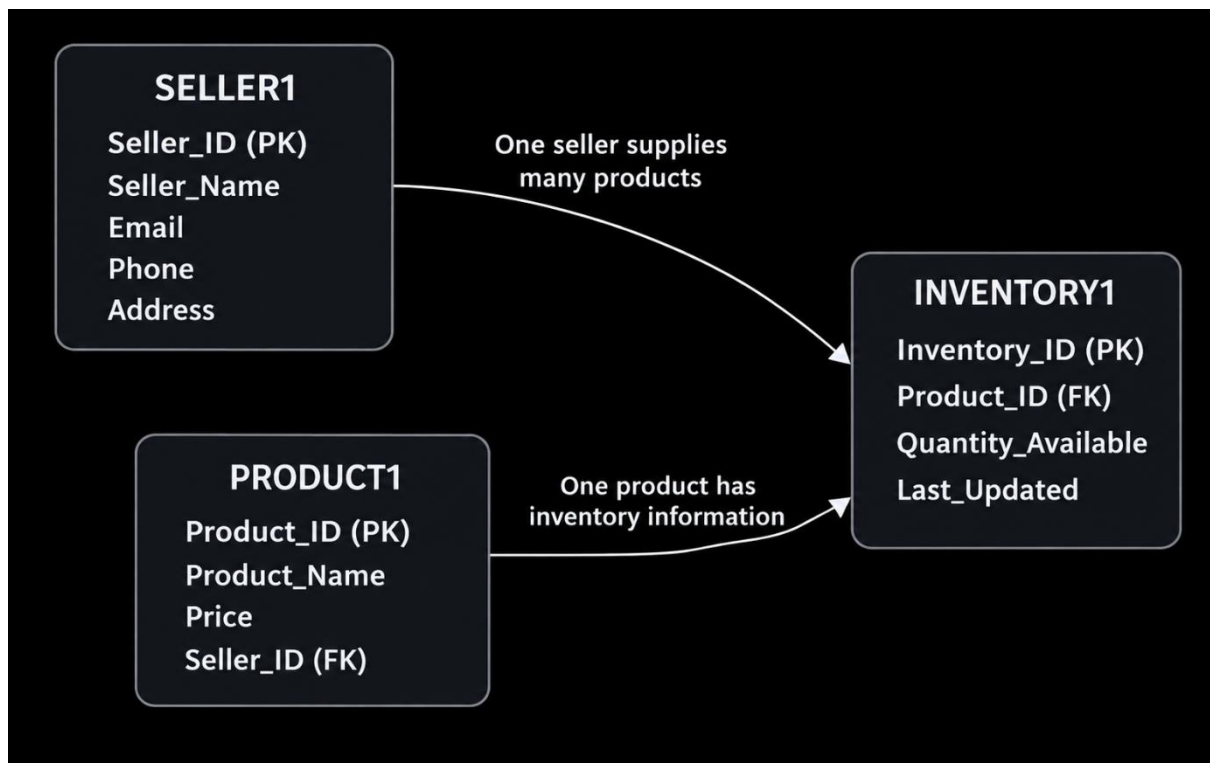
1.1 Seller

Attribute	Key	Description
Seller_ID	Primary Key (PK)	Unique ID of the seller
Seller_Name	-	Seller's business name
Email	-	Seller's email address
Phone	-	Seller's contact number
Address	-	Seller's business address

1.2 Inventory

Attribute	Key	Description
Inventory_ID	Primary Key (PK)	Unique inventory ID
Product_ID	Foreign Key (FK)	ID of the product
Quantity_Available	-	Available quantity in stock
Last_Updated	-	Date of last stock update

2.Relationship



3. Table creation

3.1 Create Seller Table

```
SQL> CREATE TABLE SELLER1 (  
2     Seller_ID INT PRIMARY KEY,  
3     Seller_Name VARCHAR2(20),  
4     Email VARCHAR2(25),  
5     Phone VARCHAR2(10),  
6     Address VARCHAR2(50)  
7 );
```

Table created.

3.2 Create Inventory Table

```
SQL> CREATE TABLE INVENTORY1 (  
2     Inventory_ID INT PRIMARY KEY,  
3     Product_ID INT,  
4     Quantity_Available INT,  
5     Last_Updated DATE,  
6     FOREIGN KEY (Product_ID) REFERENCES PRODUCT1(Product_ID)  
7 );
```

Table created.

4. Seller Data Insertion

```
SQL> INSERT INTO SELLER1 VALUES(1, 'ABC Traders', 'abc@gmail.com', '9876543210', 'Chennai');
1 row created.

SQL> INSERT INTO SELLER1 VALUES(2, 'XYZ Stores', 'xyz@gmail.com', '9876543211', 'Bangalore');
1 row created.

SQL> INSERT INTO SELLER1 VALUES(3, 'Maya Products', 'maya@gmail.com', '9876543212', 'Vellore');
1 row created.

SQL> INSERT INTO SELLER1 VALUES(4, 'Sri Mart', 'srimart@gmail.com', '9876543213', 'Coimbatore');
1 row created.

SQL> INSERT INTO SELLER1 VALUES(5, 'Green Shop', 'green@gmail.com', '9876543214', 'Salem');
1 row created.
```

5. Inventory Data Insertion

```
SQL> INSERT INTO INVENTORY1 VALUES(1, 101, 10, DATE '2026-09-01');
1 row created.

SQL> INSERT INTO INVENTORY1 VALUES(2, 102, 25, DATE '2026-09-02')
2
SQL> INSERT INTO INVENTORY1 VALUES(2, 102, 25, DATE '2026-09-02');
1 row created.

SQL> INSERT INTO INVENTORY1 VALUES(3, 103, 0, DATE '2026-09-03');
1 row created.

SQL> INSERT INTO INVENTORY1 VALUES(4, 104, 15, DATE '2026-09-04');
1 row created.

SQL> INSERT INTO INVENTORY1 VALUES(5, 105, 0, DATE '2026-09-05');
1 row created.
```

6. Seller Product Information

6.1 Display Seller and Product Details

```
SQL> SELECT
  2     s.Seller_ID,
  3     s.Seller_Name,
  4     p.Product_ID,
  5     p.Product_Name,
  6     i.Quantity_Available
  7 FROM SELLER1 s
  8 JOIN PRODUCT1 p
  9     ON s.Seller_ID = p.Seller_ID
 10 JOIN INVENTORY1 i
 11     ON p.Product_ID = i.Product_ID;
```

SELLER_ID	SELLER_NAME	PRODUCT_ID	PRODUCT_NAME	QUANTITY_AVAILABLE
1	ABC Traders	101	Laptop	10
2	XYZ Stores	102	Mouse	25
3	Maya Products	103	Keyboard	0

SELLER_ID	SELLER_NAME	PRODUCT_ID	PRODUCT_NAME	QUANTITY_AVAILABLE
4	Sri Mart	104	Headphones	15
5	Green Shop	105	Monitor	0

7. Inventory Status

7.1 Display Available Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Quantity_Available
  5 FROM PRODUCT1 p
  6 JOIN INVENTORY1 i
  7 ON p.Product_ID = i.Product_ID
  8 WHERE i.Quantity_Available > 0;
```

PRODUCT_ID	PRODUCT_NAME	QUANTITY_AVAILABLE
101	Laptop	10
102	Mouse	25
104	Headphones	15

7.2 Display Unavailable Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Quantity_Available
  5 FROM PRODUCT1 p
  6 JOIN INVENTORY1 i
  7 ON p.Product_ID = i.Product_ID
  8 WHERE i.Quantity_Available = 0;
```

PRODUCT_ID	PRODUCT_NAME	QUANTITY_AVAILABLE
103	Keyboard	0
105	Monitor	0

8. Inventory Update

Demonstrate updating stock information

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Quantity_Available,
  5     i.Last_Updated
  6 FROM PRODUCT1 p
  7 JOIN INVENTORY1 i
  8 ON p.Product_ID = i.Product_ID
  9 WHERE i.Quantity_Available = 0;
```

PRODUCT_ID	PRODUCT_NAME	QUANTITY_AVAILABLE
105	Monitor	0

LAST_UPDA
05-SEP-26

```
SQL> UPDATE INVENTORY1
  2 SET Quantity_Available = 30,
  3     Last_Updated = SYSDATE
  4 WHERE Product_ID = 103;
```

1 row updated.

9. Inventory Status Reports

9.1 Complete Inventory Report

```
SQL> SELECT
  2     i.Inventory_ID,
  3     s.Seller_Name,
  4     p.Product_Name,
  5     p.Price,
  6     i.Quantity_Available,
  7     CASE
  8         WHEN i.Quantity_Available > 0 THEN 'Available'
  9         ELSE 'Unavailable'
 10     END AS Stock_Status,
 11     i.Last_Updated
 12 FROM INVENTORY1 i
 13 JOIN PRODUCT1 p
 14 ON i.Product_ID = p.Product_ID
 15 JOIN SELLER1 s
 16 ON p.Seller_ID = s.Seller_ID
 17 ORDER BY i.Inventory_ID;
```

INVENTORY_ID	SELLER_NAME	PRODUCT_NAME	PRICE	QUANTITY_AVAILABLE	STOCK_STATU	LAST_UPDA
1	ABC Traders	Laptop	55000	10	Available	01-SEP-26
2	XYZ Stores	Mouse	500	25	Available	02-SEP-26
3	Maya Products	Keyboard	1200	30	Available	17-SEP-26
4	Sri Mart	Headphones	2500	15		
5	Green Shop	Monitor	15000	0		

9.2 Available and Unavailable Product Count

```
SQL> SELECT
  2     CASE
  3         WHEN Quantity_Available > 0 THEN 'Available'
  4         ELSE 'Unavailable'
  5     END AS Stock_Status,
  6     COUNT(*) AS Total_Products
  7 FROM INVENTORY1
  8 GROUP BY
  9     CASE
 10         WHEN Quantity_Available > 0 THEN 'Available'
 11         ELSE 'Unavailable'
 12     END;

STOCK_STATU TOTAL_PRODUCTS
-----
Available          4
Unavailable         1
```

9.3 Total Stock by Seller

```
SQL> SELECT
  2     s.Seller_Name,
  3     SUM(i.Quantity_Available) AS Total_Stock
  4 FROM SELLER1 s
  5 JOIN PRODUCT1 p
  6 ON s.Seller_ID = p.Seller_ID
  7 JOIN INVENTORY1 i
  8 ON p.Product_ID = i.Product_ID
  9 GROUP BY s.Seller_Name
 10 ORDER BY s.Seller_Name;

SELLER_NAME          TOTAL_STOCK
-----
ABC Traders          10
Green Shop            0
Maya Products        30
Sri Mart             15
XYZ Stores           25
```

9.4 Products with Zero Stock

This is particularly useful for demonstrating unavailable products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Quantity_Available,
  6     CASE
  7         WHEN i.Quantity_Available = 0 THEN 'Unavailable'
  8         ELSE 'Available'
  9     END AS Stock_Status
 10 FROM PRODUCT1 p
 11 JOIN INVENTORY1 i
 12 ON p.Product_ID = i.Product_ID
 13 JOIN SELLER1 s
 14 ON p.Seller_ID = s.Seller_ID
 15 WHERE i.Quantity_Available = 0;
```

```
PRODUCT_ID  PRODUCT_NAME
```

```
-----
```

```
SELLER_NAME          QUANTITY_AVAILABLE  STOCK_STATU
```

```
-----
```

```
      105 Monitor
```

```
Green Shop
```

```
0 Unavailable
```